ROBERTO TEJADA AREVALO, CV (2 PAGES)

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Research Interests:

- Formation and evolution of gas giants, ice/rock giants, sub-Neptunes, super-Earths, and rocky terrestrial planets.
- Coupled atmospheric and interior modeling.
- Exoplanet orbital dynamics coupled with interior evolution.
- Equations of state, thermodynamics, and miscibility curves for planetary interior modeling.

Education:

- Ph.D. in Astrophysical Sciences, Princeton University (2026).
- B.S. Physics, California State University, Los Angeles (2020), Summa Cum Laude.
- A.A. Physics and Mathematics, Los Angeles City College (2017).

First-Author Publications:

- **Tejada Arevalo, R.,** Burrows, A., Sur, A., Su, Y. *ORCHARD: A General Planetary Evolution Code.* 2026, ApJ, 1010, 44.
- **Tejada Arevalo, R.**, Gupta, A., Burrows, A., Zheng, D., Tang, Y., Deng, J. *Sub-Neptune Memories I: Implications of Inefficient Mantle Cooling and Silicate Rain.* 2026, ApJ, 1001, 243.
- **Tejada Arevalo, R.** *Different Inhomogeneous Evolutionary Histories for Uranus and Neptune.* 2025, ApJL, 989, L40.
- **Tejada Arevalo, R.,** Sur, A., Su, Y., Burrows, A. *Jupiter Evolutionary Models Incorporating Stably Stratified Regions.* 2025, ApJ, 979, 243.
- **Tejada Arevalo, R.,** Sur, A., Su, Y., Burrows, A. *Equations of State, Thermodynamics, and Miscibility Curves for Jovian Planet and Giant Exoplanet Evolutionary Models.* 2024, ApJS, 274, 34.
- **Tejada Arevalo, R.,** Tamayo, D., Cranmer, M. *Do Young Planets Form in Mean Motion Resonance Chains?* 2022, ApJL, 932, L2.
- **Tejada Arevalo, R.,** Winn, J., Anderson, K. *Further Evidence of Tidal Spin-up For Hot Jupiter Host Stars.* 2021, ApJ, 919, 2.

Selected Co-Author Publications:

- Kathuria, G., Bouma, L. G., Boyle, A. W., **Tejada Arevalo, R.,** et al. *An Infant Sub-Saturn in Scorpius-Centaurus.* Submitted to AAS Journals.
- Su, Y., Dewberry, J., **Tejada Arevalo, R.,** Sur, A., Burrows, A. *Saturn's Evolutionary History and Seismology: Survival of Deep Stably Stratified Regions in Evolutionary Models of Saturn Consistent with Ring Seismology.* Submitted to ApJ, arXiv:2602.02647.

- Sur, A., **Tejada Arevalo, R.,** Su, Y., Burrows, A. *Simultaneous Evolutionary Fits for Jupiter and Saturn Incorporating Fuzzy Cores.* 2025, ApJL, 980, L5.
- Sur, A., **Tejada Arevalo, R.,** Su, Y., Burrows, A. *APPLE: An Evolution Code for Modeling Giant Planets.* 2024, ApJ, 971, 104.
- Chen, Y., Burrows, A., Sur, A., **Tejada Arevalo, R.** *Jupiter Atmospheric Models and Outer Boundary Conditions for Giant Planet Evolutionary Calculations.* 2023, ApJ, 957, 36.

Open-source code:

- Tejada Arevalo et al. (2026 b). `ORCHARD`: A General Planetary Evolution Code. GitHub
- Tejada Arevalo et al. (2024). `EOS`: Equations of State for general planetary evolution. GitHub
- Blunt et al. (2019; incl. Tejada Arevalo). `orbitize!`: A Computational Framework for Modeling Exoplanet Orbits. Documentation

Selected Honors:

- Feb 2026: University of California Presidential Postdoctoral Fellowship
- May 2024: Princeton Astrophysical Sciences Dept. Equity Prize
- March 2022: Ford Foundation Predoctoral Fellowship

References:

- **Jonathan Fortney**
  Chair and Professor of Astrophysics, University of California, Santa Cruz
  jfortney@ucsc.edu
- **Adam S. Burrows**
  Professor of Astrophysical Sciences, Princeton University
  burrows@astro.princeton.edu
- **Enrico J. Ramirez-Ruiz**
  Professor of Astronomy and Astrophysics and Director of the Center for Reimagining Leadership
  raruiz@ucsc.edu
- **Josh N. Winn**
  Professor of Astrophysical Sciences, Princeton University
  jwinn@princeton.edu

Recent Talks:

- August 20, 2026: Uranus Workshop, Caltech/JPL (Talk)
- August 14, 2026: Other Worlds Laboratory Conference, UC Santa Cruz (Talk)
- March 16, 2026: AASTCS11 Conference (Talk)
- December 19, 2025: AGU 2025 Conference (Talk)
- November 25, 2025: Caltech GPS Seminar
- November 10, 2025: UC Santa Cruz Planetary Lunch Seminar
- November 5, 2025: Yale Exoplanet and Stars Seminar
- October 21, 2025: MIT EAPS Planetary Lunch Seminar
- October 17, 2025: Caltech TAPIR Seminar
- October 16, 2025: UC San Diego Stars and Planets Seminar
- October 13, 2025: Caltech Tea Talk
- September 2025: UCLA Tuesday Lunch Talk